

Computer Networks Lab 1 & 2

Lab 1: Introduction to Computer Networks & Cisco Packet Tracer

1. Objective

- To understand the basic definition of a computer network.
- To identify different network devices (nodes).
- To install and explore Cisco Packet Tracer.
- To create a simple network topology using Packet Tracer.

2. Definition of Computer Network

According to Wikipedia:

"A computer network is a digital telecommunications network which allows nodes to share resources. In computer networks, computing devices exchange data with each other using connections (data links) between nodes. These data links are established over cable media such as wires or optic cables, or wireless media such as Wi-Fi."

3. Basic Theory

A computer network consists of multiple interconnected devices called nodes. These nodes communicate with each other to share data and resources.

Common Network Nodes:

- Router – Connects different networks together.
- Switch – Connects devices within the same network.
- Firewall – Provides network security and controls traffic.
- End Devices – Clients and Servers.
- Access Point – Connects wireless devices to the network.
- Other devices may include hubs, modems, etc.

4. Required Software

- Cisco Packet Tracer (Latest Version)

5. Procedure

Part A: Installation of Cisco Packet Tracer

1. Download the Cisco Packet Tracer installation file provided by the instructor.
2. Run the setup file.
3. Follow installation steps and complete the setup.
4. Launch Cisco Packet Tracer.

Part B: Exploring Packet Tracer Interface

1. Observe the device panel (Routers, Switches, End Devices).
2. Observe the connection types (Copper straight-through, crossover, etc.).
3. Understand the workspace area where topology is created.

Part C: Understanding IP Address Configuration

Before testing connectivity, each device in a network must have a unique IP address. An IP address is a logical address used to identify a device in a network.

In this lab, students will manually assign IP addresses to PCs in Cisco Packet Tracer.

Each PC must have:

- IP Address
- Subnet Mask

For this lab:

- Network: 192.168.1.0
- Subnet Mask: 255.255.255.0

Both PCs must be in the same network to communicate.

Part D: Configuring IP Address in Cisco Packet Tracer

Follow these steps:

1. Click on the PC.
2. Go to the **Desktop** tab.
3. Click on **IP Configuration**.
4. Select **Static**.
5. Enter the IP Address and Subnet Mask as instructed.
6. Close the window.

Repeat the same steps for the second PC using a different IP address.

Part E: Introduction to Command Line Interface (CLI)

The Command Line Interface (CLI) allows users to interact with the system by typing commands.

In networking, CLI is used to:

- Check connectivity
- View IP configuration
- Troubleshoot network issues
- Monitor network status

In Cisco Packet Tracer:

1. Click on the PC.
2. Go to the **Desktop** tab.
3. Click on **Command Prompt**.
4. Type the required networking commands.

Part F: Basic Networking Commands

The following commands will be used in this lab:

1. `ipconfig`

Displays the current IP address configuration of the PC.

2. `ping`

Tests connectivity between two network devices.

3. `tracert`

Shows the path taken by packets to reach the destination.

4. `arp -a`

Displays the ARP table (IP to MAC address mapping).

5. `netstat`

Displays active network connections and statistics.

These commands help in verifying whether devices in a network are properly connected and communicating.

6. **Expected Result**

Students should understand the basic concept of computer networks, identify network devices, install Cisco Packet Tracer successfully, and create a simple network topology.

7. Lab Task

Task: Create and Test a Simple LAN in Cisco Packet Tracer

Part A – Network Topology Creation

1. Open **Cisco Packet Tracer**.
2. Add the following devices to the workspace:
 - **1 Switch**
 - **2 PCs**
3. Connect:
 - PC0 to Switch
 - PC1 to Switch

Part B – IP Address Configuration

Configure IP addresses manually:

- **PC0**
 - IP Address: 192.168.1.1
 - Subnet Mask: 255.255.255.0
- **PC1**
 - IP Address: 192.168.1.2
 - Subnet Mask: 255.255.255.0

(No Default Gateway required because no router is used.)

Part C – Network Testing Using CLI Commands

Open **Command Prompt** on each PC and perform the following commands:

1. `ipconfig`

Purpose: Displays IP configuration of the PC.

Example:

```
ipconfig
```

2. `ping`

Purpose: Tests connectivity between two devices.

From PC0:

```
ping 192.168.1.2
```

From PC1:

```
ping 192.168.1.1
```

You should receive **Reply from...** messages.

3. `tracert`

Purpose: Shows the path taken by packets.

From PC0:

```
tracert 192.168.1.2
```

Since both PCs are connected through a switch, it will show only one hop.

4. `arp -a`

Purpose: Displays ARP table (MAC address mapping).

```
arp -a
```

You should see the MAC address of the other PC after pinging.

5. `netstat`

Purpose: Displays active connections and statistics.

```
netstat
```

8. Submission Instructions

- Perform the given lab task.
- Take clear screenshots of each major step.
- Paste the screenshots in a Word file.
- Write a short explanation (2–4 lines) below each screenshot.
- Submit the soft copy to the instructor before the deadline.